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Craving for Order

After the heated session in the conference room, welcome back to the studio. Having listened in on de Troy's disquisition on the merits of destruction, I must admit that it's hard to fault his argument that much of creation in language is just a by-product of erosion. If there is anything to quibble with, it is only with the ringing overtones of completeness. By trumpeting the forces of destruction, and boasting how much they can achieve, de Troy may have given the impression that erosion can account for *all* the elaborate structures in language. But here I beg to differ. Even though the forces of destruction can explain an enormous amount, they cannot account for everything. And some of the structures that erosion alone cannot explain happen to be among the most dazzling architectures found in the world's languages.

One striking example is an edifice which I mentioned in Chapter 1, the verbal system of the Semitic languages. You may recall that the root of a Semitic verb is not a pronounceable string of consonants and vowels, like English 'twist' or 'turn', but an abstract entity which consists exclusively of consonants. Roots such as Arabic s-l-m 'be at peace' or Hebrew š-b-t 'rest' (š stands for the sound *sh*) come to life only when they are inserted into a 'template': a sequence of sounds with empty slots for the three root-consonants. The Hebrew template ○a○a○, for instance, expresses the past tense (in the third person 'he'), so when the root š-b-t is superimposed on this template, it yields šabat 'he rested' (hence 'Sabbath' – the day on which 'He rested'). When the same root is inserted into other templates, it generates various other nuances of the verb. In the template ○o○e○, for example, the root creates the form šobet 'he rests'; the template hu○○○e○ta yields hušbet a 'she was made to rest', and the template na○○○i○ gives našbit 'we will cause to rest' or 'we will bring to a standstill' (typically said by striking workers). These are

just a handful of simple examples, but as I mentioned earlier, there are many dozens of such templates, through which the Semitic languages can express every conceivable nuance of the verb.

But isn't it possible that the forces of erosion alone could have created such a system? After all, haven't we just seen that the wear and tear of erosion can create finely laced structures like the French verbal system, with its criss-crossing of endings for different persons, tenses and other nuances? Why couldn't a structure like the Semitic verb have developed in much the same way? The fact is that while erosion can create endings – reams and reams of them – the structure on display here is of an entirely different order. What makes the Semitic verbal architecture so special is not so much the sheer bulk of the templates, but rather the remarkable idea behind their design, the system of tri-consonantal roots and prefabricated vowel templates. There is just no way that erosion on its own could ever have come up with such an abstract algebraic scheme, a conceptual design of roots that cannot even be pronounced, but which are superimposed on vowel templates to produce every conceivable nuance of the verb. In fact, if there is anything in language which still seems to cry out for a conscious invention, this is surely it. For if it was not invented, how could people ever have stumbled across such an unusual idea?

And yet, there is an alternative explanation. We do not have to call on a *deus ex machina* to account for the origin of designs such as the Semitic verbal system, nor is there any need to discern the guiding hand of an architect in their construction. The following pages will try to show that it is within our grasp to understand how abstract linguistic designs could have arisen of their own accord. But if there is to be any chance of success in this enterprise, then we cannot pin all our hopes on erosion alone. We have to call on another essential element, one which the previous chapters have rather neglected.

Chapter 2 mentioned a triad of motives for language's inner restlessness: economy, expressiveness and analogy. So far, however, only the first two of these motives have received much attention: economy, which causes the erosion in sounds, and expressiveness, which results in the inflationary erosion in meaning and drives the flow of metaphors from the concrete to the abstract. The role of analogy was acknowledged only summarily (in Chapter 4), as the cognitive mechanism behind our ability

to find similarities between different domains, that is, our capacity for metaphorical thinking. Nevertheless, to understand how abstract designs can emerge in language, what will prove critical is exactly this overlooked third part of the triad, analogy, or the mind's craving for order.

This chapter thus sets out to redress the balance, and explore the power of analogy. Analogy will soon emerge as the main element of 'invention' in the course of language's evolution. Nonetheless, this type of invention does not spring from the design of any architect, nor does it follow any careful plan. The element of invention comes from thousands of spontaneous attempts by generation upon generation of order-craving minds to make sense of the chaotic world around them. And as we shall see, the force of such spontaneous innovations can sometimes accumulate to create imposing linguistic structures. By exploring the role of analogy in language change, we will also complete the survey of the central mechanisms of linguistic creation, and will thus be on course for our ultimate goal, projecting our findings onto the distant past in order to discover how the full complexity of language could gradually have evolved.

TO THINK IS TO FORGET A DIFFERENCE

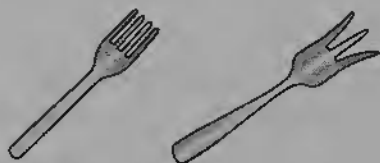
In his story 'Funes the Memorious', the Argentinian writer Jorge Luis Borges tells of a man called Funes, who lost consciousness when he was thrown off a horse and after regaining it found that he couldn't forget anything he had ever seen or heard. 'He remembered the shapes of the clouds in the south at dawn on the 30th of April of 1882, and he could compare them in his recollection with the marbled grain in the design of a leather-bound book which he had seen only once . . .' But it is because of this unusual gift that Funes is incapable of any real thought – he is simply drowning in detail:

It was not only difficult for him to understand that the generic term *dog* embraced so many unlike specimens of differing sizes and different forms; he was disturbed by the fact that a dog at 3.14 (seen in profile) should have the same name as the dog at 3.15 (seen from the front) . . . I suspect that he was not very capable of thought. To think is to forget a difference, to

generalize, to abstract. In the overly replete world of Funes, there were nothing but details, almost contiguous details.

Borges understood that the ability to pick out patterns, to draw analogies between unequal yet similar things, in short, to 'forget a difference', is at the very core of our intelligence. And the process of mastering a language is a good illustration of the role of analogy in enabling us to cope with an overwhelming amount of detail. As anyone who has tried to learn a foreign language will remember, the more order and regularity that can be picked out, the fewer the forms that need to be memorized individually. (An old German proverb says that keeping order is a crutch for those who are too lazy to search for things . . .) Were it not possible to extract any recurrent patterns from the mass of new information to be absorbed, our minds would simply be swamped by detail.

The ability to pick out patterns is not only crucial when learning a foreign language, it is just as vital to young children grappling with their mother-tongue. Babies do not imbibe their language with their mother's milk, they have to work out the whole darned system for themselves, and the mass of information they have to take in is mind-blowing. The burden becomes lighter, however, the more recurrent patterns they can identify. So it's no wonder that children act on the assumption that as much as possible in language should follow simple regular rules, hence cute errors such as 'I goed', 'my twoth birthday', 'foots', and so on. These mistakes are nothing other than perfectly sensible attempts to introduce order to corners of the language which happen to be quite messy and irregular. Sometimes, children even manage to outwit language's most basic principle of the arbitrary sign. Not content with the idea that words mean something only by convention, they find meaningful patterns in the most random of words. An oft-cited case is that of a toddling clever-clogs, who, when presented with a fork with only three prongs, studied it intently and quite naturally pronounced it to be a 'threek'.



Fork and threek

As they grow up, children gradually come to learn which areas of their language do not abide by regular rules, so that most of the errors are corrected: 'twoth' is replaced by 'second', 'foots' and 'mouses' by 'feet' and 'mice', and so on. All the same, if such errors do persist beyond childhood, they can sometimes gain ground, and ultimately overtake some well-established forms. I mentioned in Chapter 2, for instance, that the nouns *eye* and *cow* originally had irregular plurals *eyn* and *kine*. But at some stage, the 'errors' *eyes* and *cows* caught on, and eventually usurped the original forms.

While the most ear-catching mistakes are certainly those made by young children, analogical innovations can also come from grown-ups. Here is one recent example which is much more likely to have originated from the speech of teenagers than toddlers, but which is nonetheless based on very similar analogical principles. Some time in the early 1960s a new coinage made its way into British English, and quickly gained currency after featuring in the Beatles' film *A Hard Day's Night*. In one scene, a pompous advertising agent mistakes George Harrison for a 'focus group' participant, and asks for his opinion on some new designs of shirt:

AGENT: Now, you'll like these . . . They're 'fab' and all the other pimply hyperboles.

GEORGE: I wouldn't be seen dead in them. They're dead grotty.

AGENT: Grotty?

GEORGE: Yeah, grotesque.

AGENT: (to secretary) Make a note of that word . . . I think it's rather touching really.

So the adjective 'grotty' began as a groovy abbreviation of 'grotesque', a simple case of laid-back effort-saving. But at this point in the story the order-craving mind begins to make its presence felt. As the dialogue shows, the relation of 'grotty' to its forebear 'grotesque' was initially still transparent to all who used it. But before too long, the connection with 'grotesque' was forgotten and 'grotty' set itself up in its own right, as a word meaning 'dirty' or 'shabby'. And as it happens, the new word 'grotty' fitted neatly into a simple regular pattern in the language, whereby many English nouns give rise to adjectives through adding on the ending -y: